

10. (a) Dave says the distances of each leg in a long-distance triathlon are all double those in a middle-distance triathlon.

Tom says the distances of each leg in an Olympic triathlon are double those in a sprint triathlon.

Use data from **Table 1** to explain whether Dave or Tom is correct. [2]

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- (b) Piera says Jake breathed in less air per minute at 4 minutes than at 3 minutes because his breathing rate was lower.

Petula disagrees because Jake's airflow per breath was greater at 4 minutes than at 3 minutes.

Use data from **Table 2**, and the equation below it, to explain whether you agree with Piera or Petula. [2]

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- (c) Use the information in **Table 4** and equations from pages 4 and 5 to answer the following questions.

- (i) Calculate Malcolm's BMI. [2]

BMI =

- (ii) Calculate Malcolm's maximum heart rate. [1]

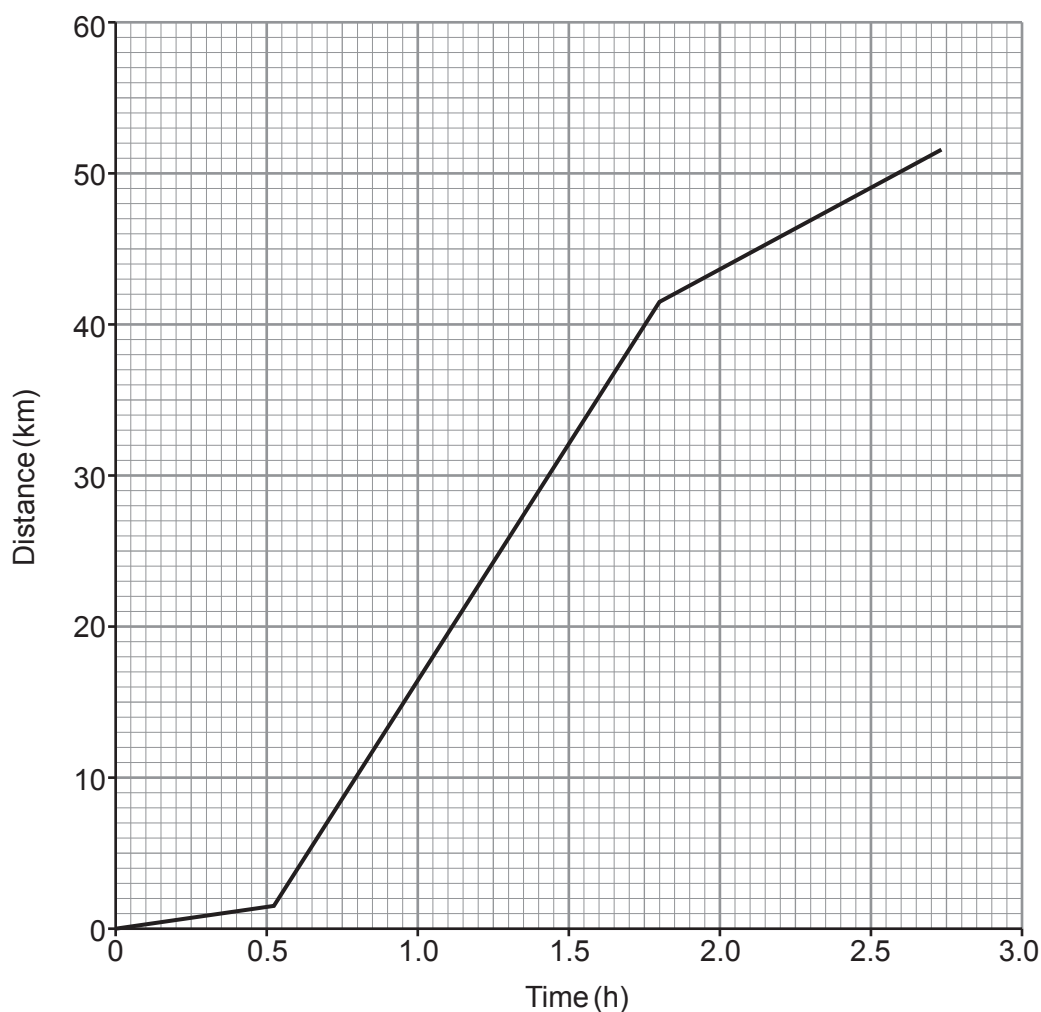
maximum heart rate = bpm



- (d) Karen wants to exercise in the aerobic zone. Use the information in **Tables 3 and 4** and on page 4 to determine the range in heart rate she should aim for. [1]

heart rate range is from bpm to bpm.

- (e) Data in the **Triathlon times** section (page 5) and **Table 1** was used to plot distance-time graphs for each of the triathlons described. There were three sections in each graph to show the swim, cycle and run legs. The graphs did not include transition times. One of the graphs is shown below.



Determine which type of triathlon is represented by the graph and give **two** reasons for your answer. [3]

Type of triathlon:

Reason 1:

Reason 2:

- (f) Use the information about Triathlon times on page 5, the information in **Table 1** and an equation on page 7 to calculate the mean speed during an Ironman triathlon. [3]

mean speed = km/h

- (g) When a triathlete mounts their bike they are travelling at 1.5 m/s. Use the information in **Table 5** and an equation on page 7 to calculate the acceleration along level ground in a cycle leg. [3]

acceleration = m/s²

- (h) It is thought that the higher the age group of the triathlete, the lower the mean speed in the cycle leg. Use data from **Graph 1** to explain whether this statement is true for all age groups. [2]

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END OF PAPER



Group

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1 H Hydrogen 1										4 He Helium 2																													
7 Li Lithium 3		9 Be Beryllium 4		23 Na Sodium 11		24 Mg Magnesium 12		39 K Potassium 19		40 Ca Calcium 20		51 V Vanadium 23		52 Cr Chromium 24		55 Mn Manganese 25		56 Fe Iron 26		59 Co Cobalt 27		59 Ni Nickel 28		63.5 Cu Copper 29		65 Zn Zinc 30		70 Ga Gallium 31		73 Ge Germanium 32		75 As Arsenic 33		79 Se Selenium 34		80 Br Bromine 35		84 Kr Krypton 36	
86 Rb Rubidium 37		88 Sr Strontium 38		89 Y Yttrium 39		137 Ba Barium 56		139 La Lanthanum 57		179 Hf Hafnium 72		181 Ta Tantalum 73		184 W Tungsten 74		186 Re Rhenium 75		190 Os Osmium 76		192 Ir Iridium 77		195 Pt Platinum 78		197 Au Gold 79		201 Hg Mercury 80		204 Tl Thallium 81		207 Pb Lead 82		209 Bi Bismuth 83		210 Po Polonium 84		210 At Astatine 85		222 Rn Radon 86	
223 Fr Francium 87		226 Ra Radium 88		227 Ac Actinium 89																																			

Key

11 B Boron 5					12 C Carbon 6					14 N Nitrogen 7					16 O Oxygen 8					19 F Fluorine 9					20 Ne Neon 10				
27 Al Aluminium 13					28 Si Silicon 14					31 P Phosphorus 15					32 S Sulfur 16					35.5 Cl Chlorine 17					40 Ar Argon 18				
70 Ga Gallium 31					73 Ge Germanium 32					75 As Arsenic 33					79 Se Selenium 34					80 Br Bromine 35					84 Kr Krypton 36				
115 In Indium 49					119 Sn Tin 50					122 Sb Antimony 51					128 Te Tellurium 52					127 I Iodine 53					131 Xe Xenon 54				
204 Tl Thallium 81					207 Pb Lead 82					209 Bi Bismuth 83					210 Po Polonium 84					210 At Astatine 85					222 Rn Radon 86				

Key

relative atomic mass

Symbol	Name
A_r	
Z	